Tristan Hoellinger

Master's degrees in Mathematics x Hybrid AI
PhD candidate in Statistical Cosmology

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I am available for postdoctoral positions from October 2026 onwards.

Academic Employment

October 2023 – September 2026 **Ph.D. Candidate**, *Institut d'Astrophysique de Paris (CNRS – Sorbonne Université)*, Paris, France
Topic: Cosmological physics with mis-modelled galaxy clustering probes.
International collaborations (full member): Aquila Consortium (2023-), Euclid consortium (2023-)
Advisors: Florent Leclercq (CR CNRS/INSU & INS2I) and Guilhem Lavaux (DR CNRS INSU; HDR).

Academic Interests

- Numerical cosmology, field-level cosmological inference,
- Implicit likelihood inference, information theory, safe machine learning,
- Deterministic and stochastic optimisation, Bayesian optimisation, control theory.

Academic Service

2023 – 2024 **Co-organiser of the weekly Friday's pre-colloquium meetings at the IAP**
Weekly informal meeting between PhD students and invited guest researchers at the IAP.

2024 **Co-animated a workshop at the Fête de la Science 2024**, *Oct 13th, 2024*
Co-animated the workshop "History of the Universe and CMB" for 11+ y.o. children.

2023 **Member of the LOC for the Aquila 2024 autumn meeting**, *Oct 21st – Oct 25th, 2024*
Bi-annual meeting of the Aquila consortium, IAP, Paris.

2023 **Member of the IAP LOC for the ML-IAP/CCA conference**, *Nov 27th – Dec 1st, 2023*
Conference on Machine Learning in astronomical surveys, IAP, Paris / Flatiron institute, New York.

Teaching

2024 - 2025 **Teaching assistant in Mathematics**, *3rd year Bachelor in Mathematics*, Sorbonne University
Differential manifolds, Stokes formulae & PDEs (38 hours), (un)constrained optimisation (38 hours).

2023 - 2024 **Teaching assistant in Mathematics**, *3rd year Bachelor in Engineering*, Sorbonne University
Distribution theory & Fourier analysis (16 hours), Fourier analysis & distribution theory (16 hours), ODEs (second year bachelor, 18 hours), numerical methods (14 hours).

Publications

in review 2025 Diagnosing systematic effects using the inferred initial power spectrum, Hoellinger and Leclercq, submitted to A&A, arXiv e-prints Dec 2024, arXiv:2412.04443.

2025 A Vorticity Confinement correction for discontinuous Galerkin schemes applied to fluid flow problems, Hoellinger, Manueco and Chapelier, 2025, IJNMHFF, 10.1108/HFF-11-2024-0854.

2023 Enhancer/gene relationships: need for more reliable genome-wide reference sets, Hoellinger et al., 2023, Front. Bioinform., Insights in Integrative Bioinformatics, PMC9999192

2020 Data-driven simulation for augmented surgery, Mendizabal et al. (incl. Hoellinger), 2020, Developments and Novel Approaches in Biomechanics and Metamaterials. Springer, hal-02538101

Invited Talks

June 27th 2025 **Cosmology & Gravitation Seminar**, *Oskar Klein Centre*, Stockholm, Sweden
Robustifying cosmological inference against model misspecification, *30 minutes talk*

March 6th 2025 **Cosmology Group Meeting**, *ICUUB — Institute of Cosmos Science*, Barcelona, Catalunya, Spain
Diagnosing systematic effects in field-based, implicit likelihood cosmological inference, *40 minutes talk*.

Contributed Talks at International Conferences

Oct 17th 2024 **X Meeting on Fundamental Cosmology**, *Universidad de Sevilla*, Sevilla, Spain
Lightening blackbox models in field-based cosmological inferences, *15 minutes talk*.

Jan 25th 2024 **Euclid France 12th Symposium**, *IP2I*, Lyon
Implicit likelihood inference in cosmology while checking for survey systematics, *12 minutes talk*.

Nov 6th 2023 **Action Dark Energy Colloquium 2023**, *LAPP*, Annecy
Implicit likelihood inference in cosmology while checking for survey systematics, *15 minutes talk*.

Other Talks

May 23rd 2025 **PhD day**, Sorbonne Université, campus Pierre-et-Marie-Curie, Paris, France
Robust cosmology with misspecified data, *5 minutes talk*.

Apr 10th 2025 **The Elbereth Conference 2025**, *by & for PhD students*, Observatoire de Paris, Meudon
Diagnosing Systematic Effects Using the Inferred Initial Power Spectrum, *10 minutes talk*.

June 7th 2024 **PhD students Seminar**, *IAP*, Paris, France
Lightening blackbox models, *40 minutes talk*.

March 29th 2024 **PhD day**, Institut d'Astrophysique de Paris
Novel methods for implicit likelihood inference in cosmology, *10 minutes talk*.

Feb 28th 2024 **The Elbereth Conference 2024**, *by & for PhD students*, IAP, Paris
Implicit likelihood inference in cosmology while checking for survey systematics, *12 minutes talk*.

Languages

English Fluent - Cambridge Linguaskill Business: C1+ (highest score, 180+/180)

French Native

Traineeships

March 2023 - July 2023 **Graduate Trainee**, *Institut d'Astrophysique de Paris*, Paris
Trainee in the "Large-scale structure and distant Universe" group at the IAP.
Advisors: Florent Leclercq & Guilhem Lavaux

February 2022 - February 2023 **Master thesis**, *ONERA - The French Aerospace Lab*, Châtillon
Research trainee in the NFLU and MSAT teams. I developed a Vorticity Confinement correction for the direct numerical simulation of Navier-Stokes equations through Discontinuous Galerkin Methods, at arbitrarily high order. Simultaneously, I implemented the method in a massively parallel solver of unsteady compressible turbulent flows. Advisors: Jean-Baptiste Chapelier & Lucas Manueco.

September 2020 - January 2022 **Junior Bioinformatics Scientist**, *Inserm*, Toulouse
In parallel to my Master's degree in Mathematics, I worked as a junior researcher in quantitative epigenomics. I studied statistical methods to predict enhancer-gene interactions in human genomes based on high-throughput data, towards application to the search of variants involved in haemochromatosis or complex genetic disorders. Advisor: Sarah Djebali.

July 2021 - August 2021 **Research Trainee — Visiting**, *Centre for Genomic Regulation*, Barcelona
Guigó lab. Automated identification and analysis of genetic variants associated with intron retention estimated from RNA-sequencing in blood cell types from the Blueprint Project. Advisor: Diego Garrido.

July 2020 **Research Trainee**, *Toulouse Mathematics Institute*, Toulouse
Stochastic calculus: theoretical study of the convergence rate of quantum non-demolition measurements, under the supervision of Tristan Benoist & Clément Pellegrini.

April 2020 - June 2020 **Research Trainee**, *IRAP - Institut de recherche en astrophysique et planétologie*, Toulouse
Theoretical study of kinetic scale plasma turbulence in the solar wind. Advisor: Philippe Louarn.

June 2019 - August 2019 **Research Trainee**, *Inria, National Institute for Research in Digital Science and Technology*
MIMESIS team. I analysed the robustness of a deep learning method for real-time biomechanical simulation, with respect to data sparsity and noises, in the context of intraoperative augmented surgery. Advisor: Andréa Mendizabal.

July 2017 **Farm Hand**, *La Ferme aux 100 Blés*, Saint-Broing-les-Moines
Agricultural operation. I used various tools (pneumatic hammer, freight elevators, riveting guns), packed flour and oil, made sourdough bread, ran a market stall and delivered farm products to clients.

Grants

2023 PhD student fellowship "Politique Scientifique" from Sorbonne Université. Value ∼100K€. This program aims to support particularly promising thesis projects, backed by high-quality applications. Only 1 fellowship awarded for Astronomy & Astrophysics in 2023.

2016 - 2020 National scholarship + merit-based grant holder during my undergrad studies (∼ 7k€/year).

International conferences, schools and workshops

2023 — 2026 **Aquila Consortium Workshops**, Paris, Oxford, Stockholm, 1 week twice a year

2025 **Ecole de Physique des Houches on Dark Universe**, Les Houches, 1 month

2024 **Rodolphe Clédassou School on Cosmology and large-scale surveys**, *Hendaye*, 2 weeks

2024 **MaxEnt 2024**, *Faculty of Economics, Ghent University*, Ghent, Belgium, 1 week
43rd International Workshop on Bayesian Inference and Maximum Entropy Methods.

2023 **Future Cosmology**, *Institut d'Études Scientifiques de Cargèse*, Corsica, 1 week

2023 **Euclid School on Cosmology and large-scale surveys**, *Ronces-les-Bains*, 2 weeks

Education

2023 - 2026 **Ph.D. Candidate**, *Sorbonne Université (Institut d'Astrophysique de Paris)*, Paris, France
Title: Cosmological physics with mis-modelled galaxy clustering probes.

2020 - 2022 **M.Sc. in Mathematics & M.Sc. in Hybrid AI**, *INSA Toulouse x N7*, Toulouse
This is a transversal curriculum leading to 2 *Diplômes d'ingénieur* (2 distinct MS degrees from French leading *grandes écoles*), built around 3 main axes: numerical mathematical modelling, statistical mathematical modelling, hybrid artificial intelligence. **Humanities are part of the curriculum**. Major scientific topics:

- deterministic and stochastic optimisation, variational data assimilation, optimal control theory,
- PDEs, finite volume and Galerkin methods, solving large linear systems, signal / image / wavelets,
- advanced statistics, reinforcement learning, advanced and physically constrained machine learning.

Master thesis: A Vorticity Confinement correction for the discontinuous Galerkin discretisation of the compressible Navier-Stokes equations: derivation and implementation in a massively parallel solver.

2019 - 2020 **Academic exchange in Physics**, *Paris-Saclay University*, Gif-sur-Yvette

I took a full year off with respect to my education in Mathematics at INSA Toulouse, in order to develop my knowledge of fundamental physics. I was enrolled in the *Magistère de Physique d'Orsay*, where I studied for one successful semester (hamiltonian mechanics, quantum mechanics, general relativity and cosmology, statistical physics). I completed my academic year with 2 traineeships in theoretical physics.

2016 - 2019 **B.Sc. in Mathematics**, *INSA Toulouse*, Toulouse

3-year intensive preparatory cycle in basic sciences ; very strong emphasis on mathematics. First part of a highly selective public 5-year curriculum leading to a *Diplôme d'ingénieur in Mathematics*.

2016 - 2017 **Propaedeutic year**, *INSA Strasbourg*, Strasbourg, *Top 10%. Transfer to INSA Toulouse.*

2013 - 2016 **High school diploma in sciences**, *Lycée Charles-Emile Freppel*, Obernai, France

Public high school. Highest Honours — Top 2% countrywide.

Practical skills

Generalities Signal/image processing, variational data assimilation, optimisation (smooth, convex, stochastic), inverse problems.

Numerical simulation Fluid dynamics, finite volume methods, Galerkin methods, particle methods for cosmology (PM, P3M, COLA).

Statistics & machine learning Parametric and non-parametric tests, stochastic processes, sampling methods, Bayesian statistics, machine learning: kernel methods, neural networks (CNNs, RNNs, GANs, PINNs), variational data-assimilation networks, wavelet scattering transforms.

HPC Solving large, ill-conditioned linear systems, multiprocessing, parallel computing (OpenMP, MPI).

Programming Python, C/C++, Fortran, Julia, R.

Misc Git, SLURM, Docker, AWS.

Leadership and Community engagement

2025 — 2026 **Member of the Committee against harassment and discriminations at the IAP**, *Paris*

2019 **Mentor of high-school students within the Equal Opportunity Program Ô Talents**, Toulouse

2019 **President of the Theatre troupe "La Catin"**, *6 months*, Toulouse

Miscellaneous

Hobbies Portrait photography, sport (running, bouldering, hiking).

Referees

Upon request.